

Ahmed-Rafik-El Mehdi BAAHMED

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Ph.D. candidate in Computer and Engineering Sciences specializing in computationally efficient Federated Learning, Edge AI, IoT systems, and distributed machine learning. Experienced in designing lightweight neural architectures, benchmarking learning systems on constrained edge-cloud infrastructures, and developing AI pipelines using TensorFlow, PyTorch, Python, Docker, Kafka, and IoT protocols. Research interests include efficient collaborative intelligence, split/federated learning, resource-aware optimization, and trustworthy AI for real-world edge deployments.

EDUCATION

Ph.D. in Computer and Engineering Sciences **ENSAM Paris / CESI LINEACT UR 7527, Strasbourg, France**
Dec. 2023 – expected Dec. 2026

- **Thesis:** *Computational Efficiency of Federated Learning Techniques on Edge-Cloud Infrastructures.*
- **Keywords:** Federated Learning, Edge AI, IoT Systems, Split Learning, Resource-Aware Optimization, Efficient Neural Networks, Distributed Machine Learning.

Master's Degree & State Engineering Diploma in Computer Systems Engineering **ESI Sidi Bel Abbes, Algeria**
Sept. 2018 – Jul. 2023

- **Master Thesis:** *Explainable Artificial Intelligence on Graph Neural Network-Based Network Intrusion Detection Systems.*
- **Engineering Thesis:** *E-GNNExplainer - Single-Instance Explanation of Edge-Classification GNNs for Network Intrusion Detection.*
- **Keywords:** Computer Systems Engineering, Graph Neural Networks, Explainable AI, Cybersecurity, Network Intrusion Detection, Deep Learning.

PROFESSIONAL EXPERIENCE

Ph.D. Researcher in Computer and Engineering Sciences **CESI LINEACT UR 7527, Strasbourg, France**
Dec. 2023 – expected Dec. 2026

- Designed and implemented experimental pipelines for evaluating Federated Learning on heterogeneous edge-cloud infrastructures, targeting smart-building and Industrial IoT applications.
- Developed Python-based machine learning workflows for model training, distributed execution, collaborative intelligence, resource monitoring, and performance benchmarking.
- Evaluated AI systems using accuracy, latency, communication cost, CPU usage, memory footprint, scalability, and knowledge-transfer efficiency.
- Implemented and tested deep learning models for edge-oriented learning scenarios using TensorFlow/Keras, PyTorch, Docker, Kafka, multiprocessing, and multithreading.
- Explored resource-aware client selection, knowledge transfer, Split Learning, and lightweight neural architectures for scalable edge intelligence.

Adjunct Lecturer in Computer Science **CESI Engineering School, Strasbourg, France**
Sep. 2024 – Apr. 2026

- Taught Artificial Intelligence, Object-Oriented Design and Programming, Computer Networks, System Administration, Arithmetic & Cryptography, Programming Paradigms, and Python for Signal Processing courses.
- Designed practical sessions and supported students in applying software engineering, AI, and systems concepts.

Deep Learning Research Intern

INSA Lyon – LIRIS Laboratory, Villeurbanne, France

Feb. – Jul. 2023

- Implemented and evaluated Graph Neural Network models for Network Intrusion Detection Systems.
- Developed explanation methods for edge-classification GNNs to improve the transparency of cybersecurity predictions.
- Built experimental workflows for model training, evaluation, and analysis using Python tools for deep learning.
- Contributed to research work on trustworthy AI, graph learning, and network security analytics.

Network Engineer Intern

HASNAOUI Group of Companies, Sidi Bel Abbes, Algeria

Sept. 2021

- Assisted with network administration, infrastructure monitoring, and technical support tasks.
- Gained practical experience with enterprise networking, system configuration, and troubleshooting.

PUBLICATIONS & SCIENTIFIC CONTRIBUTIONS

Journal Articles

- Ahmed-Rafik-El Mehdi BAAHMED et al., “HiFEL-OCKT: Hierarchical Federated Edge Learning with Objective Congruence and Multi-Level Knowledge Transfer for IoT Ecosystems,” **Internet of Things Journal**, Q1, IF 7.6, 2026.

Conference Papers

- Ahmed-Rafik-El Mehdi BAAHMED et al., “Hyperparameter Impact on Computational Efficiency in Federated Edge Learning,” **IWCMC 2024, Ayia Napa, Cyprus, May 2024**.
- Ahmed-Rafik-El Mehdi BAAHMED et al., “Using Graph Neural Networks for the Detection and Explanation of Network Intrusions,” **ECML PKDD 2023, Turin, Italy, September 2023**.
- Ahmed-Rafik-El Mehdi BAAHMED et al., “Computer Vision and Deep Learning-Based Support System for Visually Impaired People in an Outdoor Environment,” **COC’22, May 2022**.
- Ahmed-Rafik-El Mehdi BAAHMED et al., “Conception d’une Station Météo Intelligente,” **COC’21, March 2021**.

SELECTED RESEARCH AND ENGINEERING PROJECTS

Alzheimer-AI — Deep Learning for Medical Image Analysis

Nov. 2022 – Feb. 2023

- Led a team developing a computer vision and deep learning-based system for early Alzheimer’s disease detection.
- Explored multiple model architectures and image-processing approaches for medical decision-support tasks.
- Managed the project workflow, technical implementation, and experimental validation.

ArtificialEye — Computer Vision Support System for Visually Impaired People

Mar. 2022 – Jul. 2022

- Led a team developing a deep learning and computer vision-based assistive system for visually impaired users in outdoor environments.
- Contributed to model development, system design, and scientific dissemination through conference publication/presentation.

LECTIO — E-Book Web Platform

Dec. 2021 – Mar. 2022

- Managed a project team and developed a web platform for digital book access and user-oriented content

organization.

- Contributed to front-end/back-end development and project coordination.

HEALTHLINE — E-Health Web Application

Apr. 2021 – Sept. 2021

- Contributed as Web Developer and Quality Assurance Manager for a health-oriented web application.
- Participated in front-end/back-end development, software testing, feature validation, and quality control.

LEADERSHIP & COMMUNITY ACTIVITIES

Vice-President — AlphaBit Scientific Club

Jan. 2021 – Oct. 2021

- Coordinated scientific, technical, engineering, and training activities within the student community.
- Supported the organization of workshops, coding activities, and knowledge-sharing events.

Cybersecurity Instructor & Trainer — AlphaBit Scientific Club

Mar. 2021 – Oct. 2021

- Delivered introductory and applied training sessions in cybersecurity concepts and practical skills.
- Mentored students in technical learning activities and problem-solving exercises.

Developer & Organizer — AlphaBit Coding Challenge, 3rd Edition

Mar. 2021 – Jun. 2021

- Contributed to the organization and technical preparation of a national coding challenge.
- Supported event logistics, participant coordination, and challenge development.

SKILLS

Machine Learning & AI: Federated Learning, Split Learning, Edge AI, Distributed Machine Learning, Deep Learning, Explainable AI, Computer Vision, Time-Series Forecasting, Computational Intelligence paradigms.

Neural Network Architectures: CNNs, RNNs (LSTM, GRU, SimpleRNN), Attention Mechanisms, Reservoir Computing, Graph Neural Networks, Spiking Neural Networks, Deep Neural Networks, Lightweight Neural Models.

Programming & Development: Python, C/C++, JavaScript, SQL, Pascal, Git, GitHub, Docker.

Frameworks & Libraries: TensorFlow, Keras, PyTorch, scikit-learn, NumPy, Pandas, Matplotlib.

Distributed Systems & Parallel Computing: Edge-Fog-Cloud architectures, Kafka, MQTT, multiprocessing, multithreading, resource monitoring, experimental benchmarking.

IoT & Embedded Systems: Raspberry Pi, Arduino, Wireless Sensor Networks, LoRa, Sigfox, IoT cloud platforms.

Data & Analytics: Data mining/analytics, model evaluation, performance profiling, accuracy-latency-memory-CPU trade-offs analysis.

Professional Skills: Research communication, scientific writing, project coordination, teaching, teamwork, technical mentoring.

Research Interests: Efficient AI, Edge Intelligence, Federated Learning, Collaborative IoT, Trustworthy Distributed Learning.

Languages: Arabic: Native | French: Fluent | English: Fluent